

Personalized Study Planner using Retrieval-Augmented Generation (RAG) and Recommendation Techniques

M Abdurrahman Khan¹

Department of Computer Science
{ i221148 }@nu.edu.pk Instructor: Akhtar Jamil

Abstract. Recent advancements in large language models (LLMs) have revolutionized personalized learning experiences. However, existing study planners and educational chatbots are often generic and fail to adapt to individual learning styles, available time, or prior knowledge. This proposal presents a Personalized Study Planner that integrates RetrievalAugmented Generation (RAG) and recommendation algorithms to generate time-aware and goal-specific learning paths. The system retrieves relevant learning materials from course documents and produces customized study schedules using generative AI models. The proposed system aims to improve learning efficiency, enhance personalization, and reduce cognitive overload for students through adaptive planning and intelligent content retrieval.

Keywords: Retrieval-Augmented Generation · Personalized Learning · Educational Technology · Large Language Models · Recommendation Systems

1 Introduction and Problem Statement

In traditional education systems, students are often overwhelmed by the volume of available study material and lack a structured approach to achieve their learning goals efficiently. Manual study planning fails to account for individual differences such as time constraints, prior knowledge, and topic difficulty.

Recent research in Retrieval-Augmented Generation (RAG) demonstrates the effectiveness of combining external knowledge retrieval with generative models for producing more factual, contextually relevant responses. Studies such as Lewis et al. [4] and Gao et al. [3] show that RAG improves factual accuracy by grounding responses in retrieved evidence. Fan et al. [2] provide a comprehensive survey on how RAG meets LLMs, while Zhao et al. [10] explore RAG applications for AI-generated content. However, most RAG-based educational assistants focus only on question answering, not personalized learning path generation.

The proposed project aims to bridge this gap by developing a Personalized Study Planner that uses RAG to retrieve contextually relevant study content

and integrates a recommendation logic to adapt the learning plan based on user-defined parameters—such as available time, target topics, and proficiency level. Recent work in educational technology has shown promising results: Dong et al. [1] developed an adaptive AI tutor using knowledge graph-enhanced RAG, Swacha et al. [8] surveyed RAG chatbots for education, and Rao et al. [7] applied RAG for MOOC recommendation. Additionally, Wang et al. [9] explored crafting personalized agents through retrieval of personal memories, while recent advances by Linders and Tomczak [5] and Maragheh et al. [6] demonstrate the potential of knowledge graph-extended RAG and agentic RAG for personalized recommendations.

2 Objectives

The main objectives of this project are:

1. To design and implement a Retrieval-Augmented Generation (RAG) framework that retrieves relevant educational content from course-related documents (PDFs, notes, and textbooks).
2. To build a recommendation layer that sequences and prioritizes topics according to user goals and available time.
3. To generate a personalized, coherent study plan using a Generative LLM (e.g., GPT-4, Llama 3, Mistral).
4. To evaluate the system using both quantitative metrics (retrieval accuracy, ROUGE, BLEU) and qualitative feedback (student satisfaction and usefulness).
5. To conduct a comparative analysis of different embedding models and retrieval strategies (FAISS vs. Chroma, MiniLM vs. OpenAI Embeddings).

3 Proposed Methodology

The proposed architecture follows a modular RAG pipeline integrated with recommendation logic:

3.1 Step 1: Data Collection and Preprocessing

Course-related materials (lecture notes, PDFs, and online resources) will be gathered, cleaned, and converted into text format. Text will be divided into semantically coherent chunks (300–500 tokens).

3.2 Step 2: Embedding and Vectorization

Each text chunk will be embedded using pre-trained models such as all-MiniLM-L6-v2 or OpenAI text-embedding-ada-002. The resulting embeddings will be stored in a vector database (FAISS or Chroma) for efficient similarity-based retrieval.

Title Suppressed Due to Excessive Length

3.3 Step 3: Context Retrieval

Given a user query (e.g., “I have two weeks to learn the basics of Generative AI”), the retriever fetches the most relevant concepts, tutorials, and topics. Retrieval performance will be fine-tuned by adjusting chunk size and top-k parameters.

3.4 Step 4: Study Plan Generation

The retrieved context is passed to a large language model (LLM) through a structured prompt. The LLM generates a personalized, time-aware study plan that distributes topics based on difficulty, dependencies, and user time availability. A lightweight recommendation engine will rank topics using rule-based or collaborative filtering techniques.

3.5 Step 5: Evaluation

The system will be evaluated across multiple dimensions:

Retrieval Performance: Precision@k, Recall@k, and F1-score.

Generated Plan Quality: BLEU, ROUGE-L, and content coherence.

User Study: Student surveys to measure satisfaction and clarity of generated plans.

Comparative Analysis: Evaluate how different retrievers and embeddings affect accuracy and personalization quality.

4 Expected Outcomes

The expected outcomes of this research project include:

- A working AI-powered study planner capable of retrieving course-specific knowledge and generating adaptive learning schedules.
- Demonstrated improvements in contextual accuracy and personalization over standard LLM baselines.
- A comparative study of retrievers (FAISS vs. BM25) and embedding models.
- Practical insights into how RAG-based personalization can support education technology.

The final system and research results will be compiled into a paper following the Springer LNCS format.

5 Conclusion

This proposal outlines a comprehensive approach to developing a Personalized Study Planner that leverages the power of Retrieval-Augmented Generation and recommendation systems. By integrating these advanced AI techniques, we aim to create a tool that significantly enhances the learning experience for students by providing tailored, efficient, and contextually relevant study plans. The successful implementation of this system has the potential to transform educational technology and contribute valuable insights to the field of AI-assisted learning.

References

1. Dong, C., Zhang, T., Wang, Z., Liu, C., Zhang, Y.: Adaptive ai tutor using knowledge graph-enhanced retrieval-augmented generation (kg-rag). arXiv preprint arXiv:2311.17696 (2023)
2. Fan, W., Ding, Y., Ning, L., Wang, S., Li, H., Yin, D., Chua, T.S., Li, Q.: A survey on rag meeting llms: Towards retrieval-augmented large language models. arXiv preprint arXiv:2401.08912 (2024)
3. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, H.: Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2310.07514 (2023)
4. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., et al.: Retrieval-augmented generation for knowledge-intensive nlp tasks. In: Advances in Neural Information Processing Systems. vol. 33, pp. 9459–9474 (2020)
5. Linders, J., Tomczak, J.M.: Knowledge graph-extended retrieval-augmented generation for question answering. arXiv preprint arXiv:2505.10074 (2025)
6. Maragheh, R., Chen, T., Li, Y.: Arag: Agentic retrieval-augmented generation for personalized recommendation. arXiv preprint arXiv:2504.07363 (2025)
7. Rao, J., Zhang, J., Zhu, M., Zhou, Y., Chen, W., Li, J., Guo, J., Zhang, R.H.: Ramo: Retrieval-augmented generation for mooc recommendation. arXiv preprint arXiv:2407.04925 (2024)
8. Swacha, J., Bukowski, M., Woźniak, M.: Retrieval-augmented generation chatbots for education: A survey of applications. Applied Sciences 15(8), 4234 (2025)
9. Wang, Z., Xie, Z., Xu, Y., Li, J., Chen, Y., Qiao, Y., Zhang, W., Liu, T.: Crafting personalized agents through retrieval of personal memories. arXiv preprint arXiv:2406.10321 (2024)
10. Zhao, P., Zhang, H., Yu, Q., Wang, Z., Geng, Y., Fu, F., Yang, L., Zhang, W., Cui, B.: Retrieval-augmented generation for ai-generated content: A survey. arXiv preprint arXiv:2402.01429 (2024)